National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science (SEECS)

SE-200 — Software Engineering
Assignment 0.5: Project Proposal

BSCS-13 A | Group # 7

Sr No.	Name	Qalam ID
1	Maier Ali	481889
2	Aqil Ahmed	454880
3	Abdul Manan	460515
4	Muhammad Ahmed	456679

Submitted to: Dr. Gibrail Islam
Date: 4 March 2026

1. Project Overview

Land disputes are among the most common civil cases in Pakistan. The absence of a reliable, tamper-proof system for land ownership records leaves citizens exposed to fraud, long legal delays, and corrupt intermediaries. Every year, thousands of families lose property — or money — because records can be altered, forged, or simply lost.

This proposal describes LedgerLand: a browser-based land record management system built on blockchain technology. Land ownership records are stored on an immutable distributed ledger, document files are stored on a decentralized IPFS network, and ownership transfers are executed through smart contracts triggered only after biometric verification at a NADRA office. Courts are given read-only audit access to the blockchain for dispute resolution. The system is designed to serve Pakistani citizens, land revenue departments, NADRA, and the judiciary.

2. Problem Statement

Pakistan currently has no unified digital infrastructure for land ownership records. The centralized system of Punjab Land Records Authority (PLRA) creates two main problems for its operations because it depends on one location to handle all functions which also puts its entire system at risk of failure. Centralized servers store records which insiders can modify; there exists documented evidence that bribery led to changes in ownership.

Citizens selling land today must physically appear at a court, book a date weeks in advance, and sign papers before a lawyer, a process that routinely takes 30 to 60 days for a transaction that should take hours. The process operates through a network of middlemen whose interests differ from the needs of the involved parties because it moves forward at a slow pace with no visible progress.

- No secure digital record exists at the national level which can verify land ownership rights without the possibility of being tampered with.
- The land transfer process requires paper documents which make it easy for middlemen to commit fraud.
- The centralized system of PLRA prevents any citizen or court from accessing its auditing procedures.
- Lack of a single official ownership record causes ownership disputes to continue for several years.

3. Background and Related Work

The worldwide exploration of blockchain technology has proven to be a viable solution for resolving land registry issues. The Bitfury Group and the Republic of Georgia established their first blockchain land registry system in 2016 which successfully decreased fraudulent activities and accelerated property transfer processes. Sweden's Lantmäteriet conducted a similar pilot in 2017 which showed that smart contracts could replace manual transfer verification processes by using automated systems.

Propy and ChromaWay present real-world blockchain land registry systems but their current design targets developed countries which already possess digital identity systems. The unique feature of LedgerLand lies in its ability to connect with NADRA's biometric system which serves as a legitimate identity verification method that exists throughout Pakistan's existing infrastructure system while using IPFS technology to maintain land documents in a non-deletable format which remains secure even if the application backend experience security breaches.

4. Proposed Solution

LedgerLand is a web application that stores land records on an unchangeable blockchain ledger while physical documents are kept on the distributed IPFS network. Smart contracts handle ownership transfers which begin execution after users complete NADRA biometric authentication. The court system uses read-only access to the complete transaction records which helps them solve disputes more quickly and accurately.

System Components

The system diagram demonstrates that the architecture includes five components which work together to form the complete system.

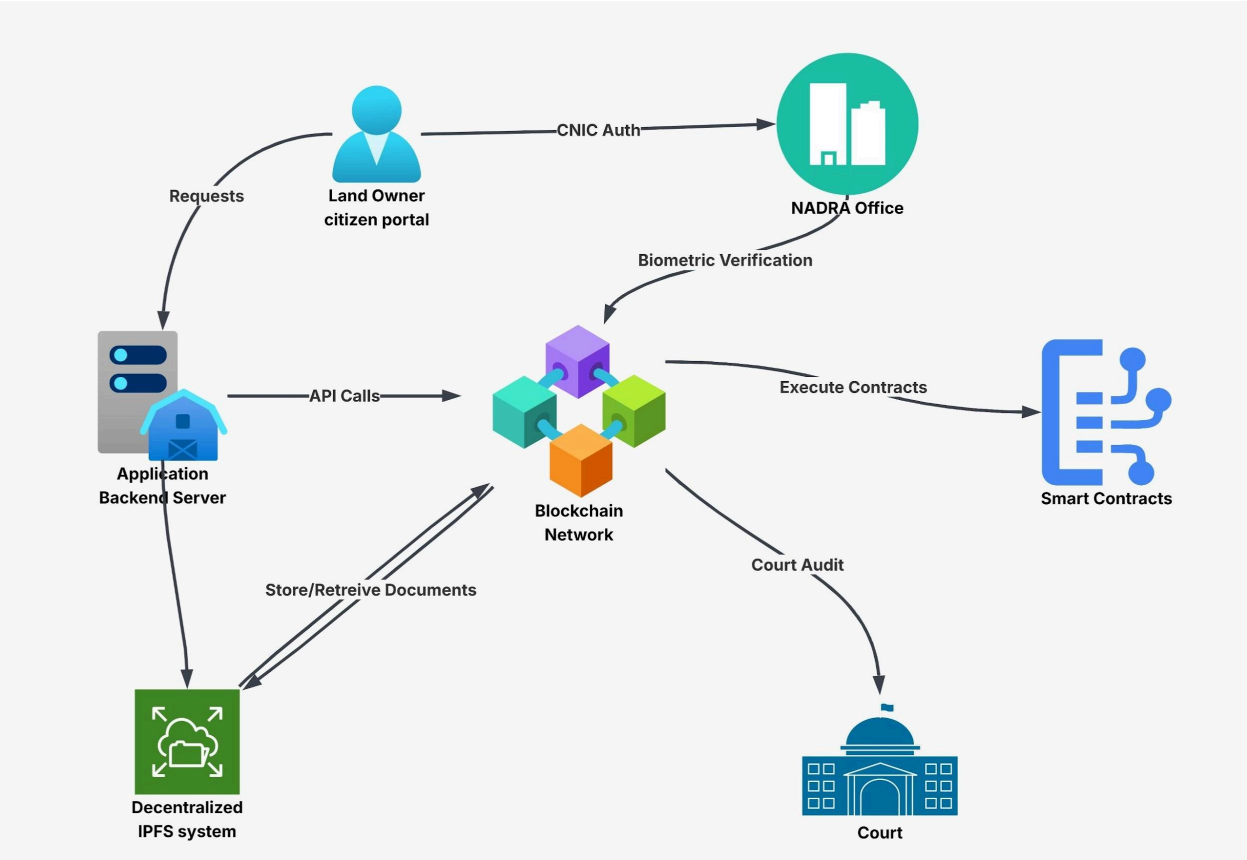
- Citizen Portal — The land owner interacts with the system through a browser-based portal. The portal starts CNIC authentication through NADRA while it forwards all user requests to the Application Backend Server.

- Application Backend Server — The central API layer. It receives requests from the citizen portal which it uses to connect with the blockchain network through API calls while it stores and retrieves documents from the decentralized IPFS system.
- Blockchain Network — The official system which maintains all records. It receives biometric verification signals from NADRA and API calls from the backend. It executes smart contracts for ownership transfers while providing audit data to the legal system.
- Smart Contracts — The blockchain network executes Solidity contracts which exist as deployed contracts. They execute automatically when all pre-conditions are met (biometric verification confirmed, both parties authenticated). The transfer becomes permanent after execution and it remains available for recording purposes.
- Decentralized IPFS System — Land ownership documents (title deeds, legal papers) are stored and retrieved from IPFS. The system maintains document integrity because no one can delete or alter documents which exist as stored records in the system.
- NADRA Office — Provides biometric verification for ownership transfers. The citizen visits a NADRA office to complete biometric authentication and the verification signal is passed directly to the blockchain network to authorize smart contract execution.
- Court — The Court Audit interface provides judges and legal authorities with read-only access to the blockchain system. This system provides judges and legal authorities with a transparent ownership history which remains unchangeable for any disputed property.
-

Core Features

- The system stores land ownership records through blockchain technology which creates permanent cryptographic signatures.
- The district-based record system allows citizens to search by their district and CNIC number to access their personal records.
- NADRA biometric integration requires physical biometric verification for all transfers, which eliminates the need for court appearances.
- Smart contract execution enables automatic ownership transfers through conditional processes which remove the need for manual work.
- IPFS document storage system stores title deeds and legal documents across multiple decentralized network nodes instead of keeping them on a single centralized server.
- The judicial system uses a court audit trail to access complete unchangeable ownership records that extend back to the beginning of time.
- The system provides citizens and government officials and courts with different levels of access through role-based access control.

System Architecture Diagram:



5. Scope of the Project

Included

- Blockchain-backed land ownership record storage and retrieval.
- District-based search with CNIC-gated citizen access.
- Simulated NADRA biometric verification for ownership transfer.
- Smart contract-based transfer execution on the blockchain.
- IPFS integration for decentralized document storage and retrieval.
- Court audit interface with read-only blockchain access.
- Admin panel for authorized government body access.
- User authentication with role-based permissions.
- Responsive browser-based interface — no installation required.

Excluded

- Live NADRA API integration — biometric step will be simulated in MVP.
- Mobile application.
- Migration of existing PLRA data.
- Agricultural or commercial land beyond residential records in MVP.

- Full dispute resolution module.

6. Tools and Technologies

Layer	Technology
Frontend	React (TypeScript), Tailwind CSS — Citizen portal and admin UI
Backend	Node.js with Express — API layer between portal and blockchain/IPFS
Blockchain	Solidity smart contracts on a permissioned Ethereum-compatible network
Document Storage	IPFS (InterPlanetary File System) — decentralized storage for land documents
Authentication	JWT for session management; NADRA biometric simulation for transfers
Smart Contracts	Hardhat development environment, OpenZeppelin contract libraries
DevOps	Docker, AWS EC2, GitHub Actions (CI/CD)
Testing	Jest (unit/integration), Hardhat test suite (smart contracts)

7. Expected Outcomes and Deliverables

The project will produce a functional web application hosted on cloud infrastructure, accessible through modern browsers without installation. Specific deliverables include:

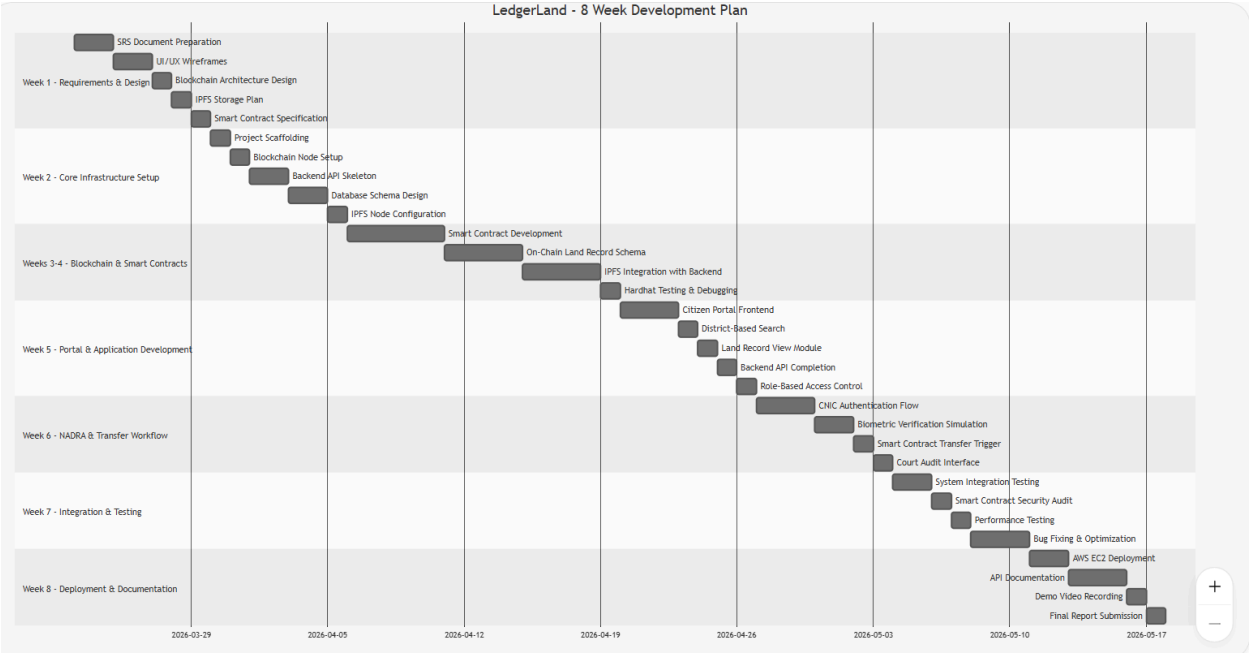
- Source code repository on GitHub with full commit history.
- Software Requirements Specification (SRS) document.
- System architecture documentation including smart contract structure and IPFS integration.
- Deployed smart contracts on a permissioned blockchain testnet.
- Test suite with minimum 70% code coverage (application and smart contracts).
- API documentation.
- Short demonstration video showing all features working end-to-end.

8. Project Timeline

The project is structured across 8 weeks, with phases progressing from requirements through deployment. The timeline is consistent with the SE-200 course schedule.

Period	Phase	Key Deliverables & Activities
Week 1	**Requirements Analysis & Design**	SRS document, UI/UX wireframes, blockchain architecture design, IPFS storage plan, smart contract specification
Week 2	**Core Infrastructure Setup**	Project scaffolding, blockchain node setup, backend API skeleton, database schema, IPFS node configuration
Weeks 3–4	**Blockchain & Smart Contract Dev**	Smart contract development and unit testing (Hardhat), land record schema on-chain, IPFS document storage integration
Week 5	**Portal & Application Development**	Citizen portal frontend (React), district search, land record view, backend API completion, role-based access

Period	Phase	Key Deliverables & Activities
Week 6	**NADRA Biometric & Transfer Workflow**	Biometric verification simulation, ownership transfer smart contract trigger, CNIC authentication flow, court audit interface
Week 7	**External Integration & Testing**	Backend–blockchain–IPFS integration testing, bug fixing, security audit of smart contracts, performance testing
Week 8	**Deployment & Documentation**	Cloud deployment (AWS EC2), API documentation, demo video recording, final report submission



9. Conclusion

Land fraud in Pakistan exists because people do not trust others who control the land. The conflicts continue for multiple years because permanent records do not exist which both citizens and courts and government officials can use as official documents. LedgerLand provides an immediate solution to this issue.

The system establishes an unchangeable record through blockchain technology while it keeps documents on IPFS and it conducts transfers through smart contracts after biometric verification and it provides courts with audit access in read-only format. The system prevents single individuals, including government clerks and corrupt officials and system administrators, from making unapproved changes to records.

The project serves academic purposes through its development of an MVP instead of being a complete production system. The system will show that people mistakenly believe technical difficulties create obstacles which prevent Pakistan from establishing a trustworthy land registry after its successful implementation. The system can be duplicated because the technology has been tested through multiple uses and Pakistan already possesses the necessary identification system through NADRA which permits operational use. The remaining barrier is institutional will, not engineering complexity.

Signatures:

Signatures (Customer):

Name	**Signature**
Aqib Ramzan	
Abdullah Khalid	
Muhammad Abdullah	
Muhammad Muqtasid	

Signature (Developer Team):

Name	**Signature**
Maier Ali	
Aqil Ahmad	
Abdul Manan	

Muhammad Ahmed	

References

[1] Graglia, J. M., & Mellon, C. (2018). Blockchain and property in 2018: At the end of the beginning. Innovations: Technology, Governance, Globalization, 12(1–2), 90–116.

[2] Nakamoto, S. (2008). Bitcoin: A peer-to-peer electronic cash system. bitcoin.org.

[3] Kempe, M. (2017). The Land Registry in the Blockchain — Testbed. Lantmäteriet, Landshypotek Bank, SBAB, Telia, ChromaWay & Kairos Future.

[4] Benet, J. (2014). IPFS — Content Addressed, Versioned, P2P File System. arXiv:1407.3561.